27 January 1976
SUPERSEDING
MIL-C-577G
20 June 1968

MILITARY SPECIFICATION

CLOTH, TWILL, NYLON, 1.6 AND 3.0 OUNCE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers nylon cloths which are suitable for coating.

1.2 Classification. The cloth covered by this specification shall be of the following types, as specified (see 6.2):

Type I	-	1.6-ounce
Type II	-	3.0-ounce
Class 1	-	Air Force use
Class 2	-	Other than Air Force use

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-P-1133 - Packaging and Packing of Synthetic Fiber Fabrics

FSC 8305

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STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

Rules and Regulations Under the
Textile Fiber Product Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

3. REQUIREMENTS

3.1 Standard sample. The dyed and finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 Fiber. The fiber shall be bright filament nylon.

3.2.2 Yarn. The yarn shall be multifilament, with 7 ± 2 turns per inch in the warp for both types I and II, and in the filling for type I.

3.3 Color. The color shall be as specified and shall match the standard sample (see 6.2 and 6.3).

3.3.1 Matching. The color shall match the standard sample under artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

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3.3.2 Colorfastness. The dyed cloth shall show fastness to dry cleaning, laundering, perspiration, crocking and light equal to or better than the standard sample. When no standard sample has been established or designated as applicable to colorfastness, the dyed cloth shall show "good" fastness to dry cleaning, laundering, perspiration and light and shall show a Munsell Value for crocking not lower than 8.5. Testing shall be as specified in 4.4.

3.4 Physical requirements. The physical requirements of the finished cloth shall be as specified in table I when tested as specified in 4.4.

TABLE I. Physical requirements

Type	Weight per sq. yd.		Yarns per inch (minimum)		Breaking strength (lbs.) (minimum)		Tearing strength (lbs.) (minimum)	
	Ounces		Warp	Filling	Warp	Filling	Warp	Filling
	Min	Max						
I	1.6	1.8	80	80	90	90	4	4
II	3.0	3.3	165	96	180	170	6	6

3.4.1 Weave. The weave of the cloth shall be as follows:

Type I - $\frac{2}{1}$ Right hand twill

Type II - $\frac{2}{2}$ Right hand twill

3.4.2 Width. The width shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of selvages.

3.5 Finish. The cloth shall be scoured, heat set and shall not be calendered. The heat setting shall be done so that the cloth meets the requirements as specified in 3.5.1.

3.5.1 Dimensional stability. The cloth shall show no appreciable distortion or puckering. There shall be no more than 3.0 percent dimensional change in either warp or filling, when tested as specified in 4.4.

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3.6 Nonfibrous materials. The total chloroform-soluble and water-soluble material content of the finished cloth shall not exceed 1.0 percent when tested as specified in 4.4.

3.6.1 Copper and manganese. The finished cloth prior to coating, shall contain not more than 0.003 percent copper nor more than 0.0015 percent manganese, when tested as specified in 4.4.

* 3.7 pH. The pH value of the finished cloth shall be no lower than 5.0 nor higher than 8.5.

3.8 Length and put-up. Unless otherwise specified (see 6.2), the finished cloth shall be furnished in continuous lengths each not less than 50 yards. Each length shall be put up in full width rolls as specified in PPP-P-1133.

3.9 Fiber identification. Each roll of cloth shall be labeled, ticketed or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

* 3.10 Face identification. The face side of the cloth shall be identified by stamping on that side with the word "Face" at each end of the roll.

3.11 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

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* 4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.2.2 Examination of the end item. Examination of the end item shall be in accordance with 4.2.2.1 through 4.2.2.3.

4.2.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects as defined in Section III of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1. No linear yard, increments of 1 yard on the measuring device of the inspection machine, from any one roll within the sample shall be penalized more than 4 points. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 28.0 points for the class 2 cloth and 20.0 for class 1.

The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 42.0 points for class 2 and 30.0 points for class 1.

If no individual roll exceeds the point level, the lot shall be acceptable, with respect to this characteristic. If one roll exceeds the point level, a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds the point level. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

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The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, rdigy or wavy cloth
Objectionable odor
Width less than specified
Finish not equal to the standard sample
Poor dye penetration, mottled, streaky or cloudy

4.2.2.2 Examination for length.

4.2.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the ticket.

4.2.2.3 Examination for shade. During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off shade, shaded side to side, side to center, or end to end, shall be cause for rejection of the entire lot represented by the sample.

* 4.2.2.4 Examination for face identification and compliance with Textile Fiber Products Identification Act. During the yard-by-yard examination each roll in the sample shall be examined for the defects listed below. The lot shall be unacceptable if two or more rolls in the sample contain one or more of the following defects:

Face identification missing from either or both ends
Not labeled or ticketed in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act

* 4.3 Examination of preparation for delivery requirements. Examination shall be made in accordance with the provisions of PPP-P-1133 to determine that packaging, packing and marking comply with section 5 requirements.

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4.4 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table II shall be followed. The physical and chemical values specified in section 3, except where otherwise specified, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 3-1/2 continuous yards, full width, of the finished cloth. All test reports shall contain the individual values utilized in expressing the final result. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more units fail to meet any requirement specified. The sample size shall be as follows:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. Test method

Characteristic	Requirement paragraph	Test method
Nylon:		
Identification	3.2.1	1530 1/
Luster	3.2.1	1/
Multifilament	3.2.2	Visual 2/
Turns per inch	3.2.2	4052
Weight	3.4	5041
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Warp	3.4	5100
Filling	3.4	5100
Tearing strength:		
Warp	3.4	5134
Filling	3.4	5134
Weave	3.4.1	Visual 2/
Colorfastness to:		
Dry cleaning	3.3.2	5620
Laundering	3.3.2	5614
Light	3.3.2	5660
Crocking	3.3.2	5651
Perspiration	3.3.2	5680

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TABLE II. Test method (cont'd)

Characteristic	Requirement paragraph	Test method
Scouring	3.5	1/
Nonfibrous material	3.6	2611
pH	3.7	2811
Copper and manganese content	3.6.1	2053
Dimensional stability	3.5.1	4.4.1

1/ Unless otherwise specified a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ One determination shall be made on each sample unit and the results reported as "pass" or "fail".

4.4.1 Test for dimensional stability. The test specimen shall be a square of cloth at least 12 by 12 inches. It shall be laid flat without tension and marked off with a square 10 by 10 inches, having sides parallel to the warp and filling directions of the cloth. The test specimen shall be placed flat on a screened tray in an air circulating oven maintained at 350°F. for a period of ten minutes. At the end of this period, the specimen shall be removed from the oven and conditioned flat on a screened tray under standard atmospheric conditions. The specimen shall then be measured for changes in dimension, and shall be visually compared with the original unheated cloth for the presence of any appreciable distortion or puckering. Any specimen showing appreciable distortion or puckering shall be cause for rejection of the lot. One specimen shall be tested from each sample unit. Change in dimension shall be reported to nearest 0.1 percent for both warp and filling. Appreciable means a change that is immediately noticeable when comparing the tested specimen with the original If closer inspection is required to make apparent a slight change, the change is not considered appreciable.

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Put-up and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put up and packaged in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

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5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1133.

6. NOTES

6.1 Intended use. The type I cloth is intended for use as base fabric for coated cloth used in the manufacture of lightweight poncho with hood and in the fabrication of capstan tubes and body bladder for high altitude flight clothing. The type II cloth is intended for use in cold weather items.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and class required (see 1.2).
(c) Color of cloth required (see 3.3).
(d) Width of cloth required (see 3.4.2).
(e) Length of roll if other than specified (see 3.8).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

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Navy - SA
Air Force - 11

Preparing activity:

Army - GL

Project No. 8305-0018

Review activities:

Army - MD
Air Force - 82
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DOCUMENT IDENTIFIER AND TITLE

MIL-C-577H - CLOTH, TWILL, NYLON, 1.6 AND 3.0 OUNCE

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A GIVE PARAGRAPH NUMBER AND WORDING

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